Lorin Urbantat

Computer Science Student @ ETH Zurich · Software Engineer @ ETH-CES

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EDUCATION

ETH Zurich, MSc Computer Science 2025 - 2027
- Expected Graduation: 2027
- Major: Data Management Systems; Minor: Machine Learning
- **Information Security Lab (5.75)**, Computational Semantics for Natural Language Processing (5.75), Advanced Formal Language Theory (5.5)

ETH Zurich, BSc Computer Science 2021 - 2024
- Graduated August 2024, GPA 5.04 (6.0 – highest)
- Parallel Programming (5.75), Numerical Methods (5.75), **Visual Computing (5.75)**, Applied Computer Architecture (6), **Human Computer Interaction (5.25)**, **Autonomous Mobile Robots (5.25)**

WORK EXPERIENCE

Information Security Lab Teaching Assistant, ETH Zurich (part-time) 2026 - Present
- Test lab exercises and host exercise sessions

Software Engineer, ETH Chair of Education Systems (part-time) 2025 - Present
- Sole engineer for a research-focused RAG chatbot serving 1,000 users. Built the application and document processing pipeline, including OCR and image descriptions, using Next.js, Convex and Microsoft Foundry.

Software Engineer, ETH Game Technology Center (part-time) 2023 - 2025
- Morph Tales Game, Unity 3D
- Developed features for an interactive Unity 3D game and worked on generative AI in game characters.
- Built a game analytics pipeline using Firebase, BigQuery and Looker Studio.

Coach, mint & pepper (part-time) 2025
- STEM and AI tutor for high school students

Calculus Teaching Assistant, ETH Zurich (part-time) 2023
- Host exercise classes, 15 - 30 students

PERSONAL PROJECTS & RELEVANT EXPERIENCE

Unikraft on Raspberry Pi 5, ETH Semester Project 2026
- Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers
- Developed a cache-safe kernel reload mechanism using OpenOCD and GDB, enabling repeated deployment without power-cycling.

Flateroo, Flat sharing platform 2024 - 2025
- React Native application with real-time chat, video streaming infrastructure
- Node backend, video transcoding, HLS streaming, PostgreSQL, fly.io

OpenDemocracy, RAG Powered Voting Advice application for elections 2024

SCION Control Plane Algorithms Test Environment, Bachelor Thesis 2024
- Built a Docker-based environment and traffic generator to evaluate SCION control-plane algorithms under configurable topologies and changing link conditions.

TLDR, Lecture Summarizer (VIScon Hackathon) 2023
- Lecture summarizer using OpenAI Whisper to create searchable lecture summaries from videos

SKILLS & INTERESTS

Languages	German (native), English (fluent), French (DELF B1)
Technologies	AArch64 Assembly, C, C++, Kernel Development, TypeScript, React, Python, Java, Git, Linux, Containerization, Unity 3D, React Native, Node.js, SQL, RAG, PCA
Interests	Biking, Sports, Cooking, Dancing, Singing